20 For all values of x

$$f(x) = 2x - 3 \quad \text{and} \quad g(x) = x^2 + 2$$

(a) Find $g(-4)$

$g(-4) = (-4)^2 + 2 = 18.$

1 mark
18

(1)

(b) Show that $gf(x) = 4x^2 - 12x + 11$

×	$2x$	-3
$2x$	$4x^2$	$-6x$
-3	$-6x$	$+9$

$g(f(x)) = (2x - 3)^2 + 2$

$= 4x^2 - 12x + 9 + 2$

$= 4x^2 - 12x + 11.$

1 mark
Compose g(f(x))

1 mark
Full expansion

(2)

(c) Solve $fg(x) = gf(x)$

1 mark
Compose and equate

$f(g(x)) = 2(x^2 + 2) - 3 = 2x^2 + 1.$

$2x^2 + 1 = 4x^2 - 12x + 11$

$(-2x^2 - 1) \qquad (-2x^2 - 1)$

$0 = 2x^2 - 12x + 10$

$(\div 2) \qquad (\div 2)$

$0 = x^2 - 6x + 5$

1 mark
Quadratic = 0

×	x	-5
x	x^2	$-5x$
-1	$-x$	$+5$

$(x - 1)(x - 5) = 0$

1 mark
Factorise to solve

$x = 1$ or $x = 5.$

1 mark
1 and 5

(4)

(Total for Question 20 is 7 marks)